

UNDERSTANDING SOCIAL MOVEMENT MESSAGING FROM A COLLECTION OF
TOPICAL IMAGES: USING MULTIMODAL SENTIMENT ANALYSIS FOR NARRATIVES
IN STRUCTURED GRAPHS

by

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that it be accepted as fulfilling the dissertation requirement.



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PREVIEW

TABLE OF CONTENTS

1	 Chapter - Introduction to Social Media News Fragmentation for Social Movements....	23
1.1	Research Problem.....	24
1.2	Research Objective	25
1.2.1	Research Questions and Hypotheses	26
1.3	Anti-Femicide Frames and Digital Fieldwork	27
1.4	Affordances of the Instagram Platform.....	28
1.5	Social Media and Fragmentation	29
1.5.1	Social Media and Social Movements	30
1.6	Social and Ethical Implications	31
1.6.1	Ethical Guidelines for Data Based on Gender Related Topics	31
1.7	Expected Research Contributions.....	33
1.8	Dissertation Outline.....	33
2	 Chapter - Exploring How Femicide is Framed in the Literature and Media	37
2.1	Femicide	39
2.2	Femicide's Narrative History in Mexico	41
2.2.1	Societal Frames	43
2.2.2	Cultural Frames.....	46
2.2.3	Social Movement Frames	50
2.2.4	Reframing and Humanizing.....	54
2.3	Femicide Frames in Digital Activism: From Global to Local Efforts	57

2.3.1	Global.....	58
2.3.2	National.....	59
2.3.3	Local.....	61
2.4	Digital Fieldwork with Citizen Data Science: Femicide at the Local Level.....	63
2.4.1	Digital Fieldwork with Ellas Tienen Nombre.....	65
2.4.2	Building the map.....	67
2.5	Reflections	71
2.5.1	How Ellas Tienen Nombre Informs this Work.....	72
3	Chapter - Conceptual Framework and Research Design.....	74
3.1	Conceptual Framework	74
3.2	Research Claim.....	75
3.2.1	Narrative Elements.....	76
3.3	Research Question	80
3.4	Visual Narrative.....	82
3.4.1	Social Media and Visual Narratives.....	84
3.4.2	Fragmented Narrative.....	85
3.5	Narrative Network.....	86
3.5.1	Image Posts within the Narrative Network	87
3.6	Research Design.....	89
3.6.1	Review of Similar Data.....	89
3.6.2	Building the Dataset.....	92
3.6.3	Collecting the Data.....	92
3.6.4	Data Annotation.....	98

3.6.5	Ethics and Privacy Concerns	99
3.7	Methodological Framework.....	100
3.7.1	Phase 1: Cluster Embedding Densities (Identifying Categories)	101
3.7.2	Phase 2: Generative M-LLM Output Comparison.....	106
3.7.3	Phase 3: Narrative Structure - Local and Global Features.....	111
3.8	Review of Methodology and Next Steps.....	116
4	Chapter - Phase 1: Cluster Embedding Densities to Identify Categories.....	118
4.1	Embedding Clusters within a Collection of Topical Images.....	120
4.1.1	Clustering	121
4.1.2	Embedding Extraction	121
4.1.3	K-Means Exploration	125
4.1.4	Parameter Tuning with HDBSCAN.....	131
4.1.5	Evaluating All Model Embeddings.....	138
4.2	Human in the Loop Evaluation: Inductive Content Analysis and Annotation.....	151
4.2.1	Inductive Content Analysis: A Systematic Approach.....	153
4.2.2	Content Analysis: Reviews of Image Coding.....	158
4.2.3	CLIP Images Representing each Cluster.....	175
4.2.4	BLIP-2 Images Representing each Cluster.....	190
4.2.5	BLIP-2 Cluster Discussion	219
4.3	Discussion	221
4.3.1	Conclusion: Network labels	223
5	Chapter - Phase 2: Comparing Generated Outputs from Multimodal LLMs	224
5.1	Deep Learning and LLM Classification Literature	225

5.2	Ground Truth Data	227
5.2.1	Label Categorization	228
5.3	Multimodal Large Language Models	231
5.3.1	Evaluation of the M-LLMs	233
5.3.2	Review Outcome for M-LLMs.....	235
5.3.3	Llama 3.2 Vision	236
5.3.4	Llama 4 Scout.....	245
5.3.5	Phi-4	254
5.3.6	Qwen 2.5-VL (Vision Language)	263
5.3.7	CLIP.....	272
5.3.8	Evaluation Metrics Discussion.....	279
5.4	Semantic Evaluation - Generated Descriptions	280
5.4.1	Semantic Similarity Analysis	282
5.5	Discussion and Findings.....	309
6	 Chapter - Phase 3: Narrative Structure Using Local and Global Features	311
6.1	Introduction	311
6.2	Narrative Structure	312
6.2.1	Review of the Literature: Network Structures.....	313
6.2.2	Review of the Literature: Multimodal classification	314
6.3	Network for the Research Claim.....	316
6.3.1	Global Structure	316
6.3.2	Local Features	317
6.4	Graph Creation – Local Features	318
6.4.1	Local Features	318

6.4.2	Graph Creation.....	319
6.5	Analyzing Global Features	353
6.5.1	Community Detection (Semantic Similarity in Graphs)	356
6.5.2	Graph Creation.....	357
6.5.3	GAT-VAE: Three Semantic Graphs.....	361
6.5.4	Evaluation of VAE Embedding Clusters	365
6.5.5	Semi-Supervised Classification: Evaluating the Annotated Graphs	376
6.6	Discussion	382
6.7	Future Work	384
7	<i>Chapter - Discussion and Future Work</i>	385
7.1	Conclusion	385
7.2	Discussion	388
7.2.1	Anti-Femicide Framing in the Media	388
7.2.2	Research Claim	389
7.3	Future Work	398
8	<i>Bibliography</i>	401

LIST OF FIGURES

1	 Chapter - Introduction to Social Media News Fragmentation for Social Movements	23
2	 Chapter - Exploring How Femicide is Framed in the Literature and Media	37
	Figure 2.1: Default Map View with Search Box.....	69
	Figure 2.2: Information with Name and image.....	70
	Figure 2.3: No Name Information	71
3	 Chapter - Conceptual Framework and Research Design.....	74
	Figure 3.1: Example of local features representing the semantic relationships between images.....	78
	Figure 3.2: Global features for the narrative network.....	79
	Figure 3.3: Community Guideline Block.....	97
	Figure 3.4: Schematic representation of the interactive machine learning process described by Mosqueira-Rey et al, 2022	106
4	 Chapter - Phase 1: Cluster Embedding Densities (Identifying Categories)	118
	Figure 4.1: Blocked images unable to be downloaded, but still counted by the scraper	122
	Figure 4.2: Silhouette Analysis of K-Means Clustering. Pre-set Cluster outputs from 2,3,4,5,6,7,8,10,12 showing cluster centroids and corresponding silhouette scores. ...	131

Figure 4.3: HDBSCAN Clusters for minimum cluster size, 40., silhouette score = 62%	
.....	134
Figure 4.4: HDBSCAN Clusters for minimum cluster size, 30. Silhouette score = 25%	
.....	135
Figure 4.5: Visualization of UMAP clusters on ResNet50 embeddings with tuned parameters, Nearest Neighbor = 25, Minimum Distance = .01.....	143
Figure 4.6: Visualization of UMAP clusters on CLIP embeddings with tuned parameters, Nearest Neighbor = 25, Minimum Distance = .01.....	144
Figure 4.7: Visualization of UMAP clusters on BLIP-2 embeddings with tuned parameters, Nearest Neighbor = 25, Minimum Distance = .01.....	145
Figure 4.8: Distribution of cluster sizes for the models CLIP, BLIP-2, and ResNet50	
.....	149
Figure 4.9: Distribution of the average probability for each point within a cluster, showing overall confidence levels for each model output.....	150
Figure 4.10: Example of max and min probabilities of first ResNet50 cluster	158
Figure 4.11: Number of Hashtags found in all ResNet50 clusters.....	160
Figure 4.12: Max and Min outlier based on probability from the ResNet50 model...	164
Figure 4.13: Image Representing Cluster 0 found with #FuimosTodas	165
Figure 4.14: Image Representing Cluster 1 found with #violenciamachista.....	166
Figure 4.15: Image Representing Cluster 2 found with #violenciadegenero.....	167

Figure 4.16: Image Representing Cluster 3 found with #8M.....	169
Figure 4.17: Image Representing Cluster 4 found with #ellasnomerepresentan	170
Figure 4.18: Image Representing Cluster 5 found with #violenciadegenero.....	171
Figure 4.19: Image Representing Cluster 6 found with #ellasnomerepresentan	172
Figure 4.20: ResNet50 cluster images posted throughout time	174
Figure 4.21: All hashtags found in the CLIP clusters, annotated and analyzed with the coding schema.....	176
Figure 4.22: Image Representing outlier group -1, found with #UnDi-aSinNosotras	180
Figure 4.23: Image Representing cluster 3, found with #FuimosTodas	182
Figure 4.24: Image Representing cluster 4, found with #FuimosTodas	183
Figure 4.25: Image Representing cluster 5, found with #25N.....	184
Figure 4.26: Image Representing cluster 6, found with #UnDi-aSinNosotras	185
Figure 4.27: Image Representing cluster 7, found with #niunamenos.....	186
Figure 4.28: Image Representing cluster 8, found with #colectivoyositecreo	187
Figure 4.29: CLIP model's posting timeline trajectory for all sample images annotated.	189
Figure 4.30: All hashtags found in the CLIP clusters, annotated and analyzed with the coding schema.....	191
Figure 4.31: Image Representing outlier group -1, found with #8M	196

Figure 4.32: Image Representing cluster 3, found with #noestamostodas.....	197
Figure 4.33: Image Representing cluster 5, found with #ellasnomerepresentan	198
Figure 4.34: Image Representing cluster 7, found with #yotecreo	200
Figure 4.35: Image Representing cluster 8, found with @abogadafemina	201
Figure 4.36: Image Representing cluster 9, found with @abogadafemina	202
Figure 4.37: Image Representing cluster 10, found with #mexicofeminicida	203
Figure 4.38: Image Representing cluster 13, found with @brujamixteca.....	204
Figure 4.39: Image Representing cluster 14, found with @UnDi-aSinNosotras.....	205
Figure 4.40: Image Representing cluster 15, found with @noestamostodas.....	206
Figure 4.41: Image Representing cluster 16, found with #femicidioemergencianacional	207
Figure 4.42: Image Representing cluster 17, found with @abogadafemina	208
Figure 4.43: Image Representing cluster 18, found with #8M.....	209
Figure 4.44: Image Representing cluster 19, found with @abogadafemina	210
Figure 4.45: Image Representing cluster 20, found with #yotecreo	211
Figure 4.46: Image Representing cluster 21, found with @niunamashmo	212
Figure 4.47: Image Representing cluster 22, found with @abogadafemina	214
Figure 4.48: Image Representing cluster 23, found with #ellasnomerepresentan	215

	Figure 4.49: Image Representing cluster 24, found with @abogadafemina	216
	Figure 4.50: Image Representing cluster 25, found with @abogadafemina	217
	Figure 4.51: Image Representing cluster 26, found with #8M.....	218
	Figure 4.52: Image Representing cluster 27, found with #yotecreo	219
	Figure 4.53: BLIP-2 model's posting timeline trajectory for all sample images annotated	221
5	Chapter - Phase 2: Comparing Generated Outputs from Multimodal LLMs.....	224
	Figure 5.1: Value counts for each of the classes annotated in label studio (Ground Truth)	229
	Figure 5.2: Llama 3.2 Vision, Label 1 Confusion Matrix.....	242
	Figure 5.3: Llama 3.2 Vision, Label 2 Confusion Matrix.....	244
	Figure 5.4: Llama 4 Scout, Label 1 Confusion Matrix.....	251
	Figure 5.5: Llama 4 Scout, Label 2 Confusion Matrix.....	253
	Figure 5.6: Phi-4, Label 1 Confusion Matrix	260
	Figure 5.7: Phi-4, Label 2 Confusion Matrix	262
	Figure 5.8: Qwen 2.5-VL, Label 1 Confusion Matrix.....	269
	Figure 5.9: Qwen 2.5-VL, Label 2 Confusion Matrix.....	271
	Figure 5.10: CLIP, Label 1 Confusion Matrix.....	277
	Figure 5.11: CLIP, Label 2 Confusion Matrix.....	278

Figure 5.12: Llama 4 Scout’s generated image label representing, “an illustration or cartoon”	285
Figure 5.13: Llama 4 Scout’s generated image label representing, “a sign(s) or banner(s)”	287
Figure 5.14: Llama 4 Scout’s generated image label representing, “solidarity”	288
Figure 5.15: Llama 4 Scout’s generated image label representing, “a person or selfie”	290
Figure 5.16: Llama 4 Scout’s generated image label representing, “informational”. 291	
Figure 5.17: Llama 4 Scout’s generated image label representing, “personal belongings or objects”	292
Figure 5.18: Llama 4 Scout’s generated image label representing, “a digital image with text in spanish”	294
Figure 5.19: Llama 4 Scout’s generated image label representing, “a digital image with text in english”	295
Figure 5.20: Llama 4 Scout’s generated image label representing, “protest”	297
Figure 5.21: Llama 4 Scout’s generated image label representing, “a small group of people”	299
Figure 5.22: Llama 4 Scout’s generated image label representing, “an image of a woman’s face with text”	301

Figure 5.23: Llama 4 Scout's generated image label representing, "an image of a mans face with text"	303
Figure 5.24: Llama 4 Scout's generated image label representing, "no label"	305
Figure 5.25: Llama 4 Scout's generated image label representing, "a digital image with text in portugeuse"	306
Figure 5.26: Llama 4 Scout's generated image label representing, "a digital image with text in italian"	308
6 Chapter - Phase 3: Narrative Structure Using Local and Global Features	311
Figure 6.1: Network of Label Relations from Llama4 Scout Generated Labels. There are 16,567 nodes and 55,337,973 edges. The density score of the network is 0.4033 ..	322
Figure 6.2: Degree Centrality Measures for Network.....	325
Figure 6.3: 10 Images representing highest Degree centrality measures; in no particular order.....	330
Figure 6.3: Histogram of Eigenvector Centrality Measures.....	333
Figure 6.4: Eigenvector centrality focused graph for Llama 4 Scot generated labels	334
Figure 6.5: 10 Images representing highest Eigenvector centrality measures; in no particular order.....	338
Figure 6.6: Histogram of Closeness Centrality measures	340
Figure 6.7: Closeness Centrality Measures graph for Llama 4 Scot generated labels	341

Figure 6.9: Graph showing the connections that support the interrelations between degree and closeness centralities within the network.	342
Figure 6.10: Histogram of betweenness centralities for the Llama4 Scout label network graph	344
Figure 6.11: Betweenness Centrality Plot for Llama4 Scout generated image labels	345
Figure 6.12: Modularity Graph with Leidan Partitioning. Represents the four communities found within the network. Modularity score = 0.2264.....	352
Figure 6.13: Threshold comparisons for Cosine Similarity Distribution of the Graph with Generated labels and descriptions. A threshold comparison of 0.60 and 0.65 ...	358
Figure 6.14: Semantic Structure of G-ALDC showing a sparse continuous narrative trajectory with colors representing 13 labels used to annotate the images.	370
Figure 6.15: Semantic Structure of G-LD showing a continuous narrative trajectory with colors representing the 17 labels generated by the Llama4 Scout Model.	372
Figure 6.16: Semantic Structure of G-LDC showing a continuous narrative trajectory with colors representing the 17 labels generated by the Llama4 Scout Model.	373
Figure 6.17: UMAP Visualization of the Semantic Structure of G-ALDC compared to Cora and Flickr Benchmarks.	380
7 Chapter - Discussion and Future Work.....	385

LIST OF TABLES

1	 Chapter - Introduction to Social Media News Fragmentation for Social Movements	23
2	 Chapter - Exploring How Femicide is Framed in the Literature and Media	37
3	 Chapter - Conceptual Framework and Research Design.....	74
	Table 3-1: Hashtags and Accounts Scraped off Instagram with the bot	93
4	 Chapter - Phase 1: Cluster Embedding Densities to Identify Categories.....	118
	Table 4-1: Minimum Cluster Size Parameter Tuning on ResNet50 Embeddings	132
	Table 4-2: Minimum Cluster Size + Minimum Sample Size: Parameter Tuning	136
	Table 4-3: Comparison of Evaluation Metrics on Minimum Cluster Parameters	137
	Table 4-4: Default UMAP Parameter Evaluation Metrics	139
	Table 4-5: UMAP Parameter Tuning for BLIP-2 Embeddings, n_neighbor and min_distance.....	140
	Table 4-6: UMAP Density Evaluation Metrics Default and Tuned Parameters - Nearest Neighbor = 25, Minimum Distance = .01	141
	Table 4-7: Review of High Confidence Points per Model	146
	Table 4-8: Outlier Review per Model.....	147
	Table 4-9: Coding dictionary for Content Analysis.....	154
	Table 4-10: Hashtag and Account Usage per ResNet50 Cluster. Size of cluster is also noted	161

	Table 4-11: Hashtag and Account Usage per CLIP Cluster. Size of cluster is also noted	176
	Table 4-12: Hashtag and Account Usage per CLIP Cluster. Size of cluster is also noted	191
5	 Chapter - Phase 2: Comparing Generated Outputs from Multimodal LLMs.....	224
	Table 5-1: Descriptions and Labels that Represent Local Features	229
	Table 5-2: Overall review of Evaluation Metrics for all Models	235
	Table 5-3: Evaluation Metrics for Llama 3.2 Vision	237
	Table 5-4: Accuracy Metrics for Top Two Labels Generated by Llama 3.2 Vision Model.....	239
	Table 5-5: Evaluation Metrics for Llama 4 Scout.....	246
	Table 5-6: Accuracy Metrics for Top Two Labels Generated by Llama 4 Scout Model	248
	Table 5-7: Evaluation Metrics for Phi 4	255
	Table 5-8: Accuracy Metrics for Top Two Labels Generated by the Phi-4 Model	257
	Table 5-9: : Evaluation Metrics for Qwen 2.5-VL.....	264
	Table 5-10: Accuracy Metrics for Top Two Labels Generated by the Qwen 2.5-VLModel.....	266
	Table 5-11: Evaluation Metrics for CLIP.....	273

	Table 5-12: Accuracy Metrics for Top Two Labels Generated by the CLIP Model..	274
	Table 5-13: BERTScorer and Sentence Transformer.	283
6	 Chapter - Phase 3: Narrative Structure Using Local and Global Features	311
	Table 6-1: Representation of linked images based on labels with weights	320
	Table 6-2: Centrality Measure Correlation Matrix.....	323
	Table 6-3: Summary Statistics of Degree Centrality Measures for Nodes	325
	Table 6-4: Image nodes with top degree centrality measure	326
	Table 6-5: Summary Statistics of Eigenvector Centrality Measures for Nodes.....	332
	Table 6-6: Image nodes with top eigenvector centrality measure	335
	Table 6-8: Summary Statistics of Betweenness Centrality Measures for Nodes	343
	Table 6-9: Betweenness Centrality Measures for images based on Llama4 Scout Labels	345
	Table 6-10: Community sizes of various modules in the network.....	352
	Table 6-11: Graph Evaluation for G-LD Thresholds 0.60 and 0.65	358
	Table 6-12: Comparison of Evaluation Metrics and Graph Density Statistics for the three graphs	367
7	 Chapter - Discussion and Future Work.....	385
8	Bibliography	401

ABSTRACT

This dissertation research explores how fragmented social media messaging about a particular social movement (the anti-feminicide movement) is represented as an overall narrative. Social movements have been using social media to share their goals, values, and grievances with the world for the past decade, which spread throughout platforms usually in visual form, and across various perspectives. This messaging is sprinkled in with other news and cultural events that fail to provide a full narrative of what is actually happening. I analyze the visual and textual narrative representations of the online anti-feminicide movement in Mexico using narrative structure, network science, visual analysis, and a combination of multi-modal computer vision / natural language processing tools. The codebase and other resources can be found here: https://github.com/lwdozal/Narrative_Network_Pipeline

The overall research is a mixed-methods research project that answers the question, how can we identify a narrative structure from a collection of topical images using computational methods. It provides an in-depth review of the feminicide history in Mexico and its related anti-feminicide movement. It provides detailed descriptions of how the media has been framed since the 1990s using: Societal, Cultural, Social Movement, and Humanization. The analysis is completed in a three-step pipeline to build a narrative structure representative of the local semantics of image groups and an overall global summarization of the narrative structure as a whole.

This first step pulls feature embeddings from a collection of 16,657 Instagram images pertaining to the anti-feminicide movement in Mexico; and clusters them to identify common vector embedding groups within the data. Human-in-the-Loop (HITL) content analysis is applied to provide a qualitative understanding of the images within each cluster group and gain insight on the classification aspects of the images for downstream tasks. The next step takes these images and runs them through foundational vision transformer models, along with generative Multimodal LLMs (M-LLMs). These models are tasked with generating two labels identified in the first set, and a description of each image. The outputs are compared and analyzed using multi-label classification metrics for machine learning models along with HITL discourse analysis of the categorization process. Step three uses the top labels created from the model with the best overall quantitative and HITL discourse evaluation metrics to build a network graph representing the various stages of semantics found within the image and its attributes 1) labels 2) descriptions and original post comments. The graph is analyzed to identify structural patterns of the network using traditional centrality measures and a modularity algorithm (Leiden).

The process identified three local features, Narrator, Action, and Subject and found that Narrator and Subject features were typically tied together as label attributes while the Action feature stood alone. Showing how the narrative of the social movement messages is structured by the generated labels. The graph is of labels, generated descriptions and original post comments

analyzed to identify structural sentiment patterns of the network using traditional centrality measures and modularity models (Leiden algorithm) within and across the network. The overall graph structure is compared to benchmark datasets in a Graph Attention Network (GAT) model that uses a variational autoencoder to compare classifications of the collected annotated image data. In the results and discussion section of the dissertation, these quantitative findings are used in conjunction with a theoretical analysis of how the media frames the anti-femicide movement in Mexico and implements an analysis review of the network's local and global semantics creating narrative structure for the collection of topical images.

PREVIEW

1 | Chapter - Introduction to Social Media News

Fragmentation for Social Movements

As a global society living in the digital age of rapid information transfer, it's difficult to know what is actually going on around the world, or even in our backyard. We are social media content consumers who receive small bits of information that provide a story from one or multiple perspectives. The affordances of our preferred platforms dictate the information we get, but similar to news cycles, seemingly important information is spread rapidly and then pivots to a new more attention-grabbing story (Mourão, Brown, 2021). Unless a viewer does extra research, is honed in on the background, or understands the full context of a social media news event, it is difficult to understand the nuances and supporting implications of a viral news event.

This was my experience in August of 2019 as anti-femicide protests crowded Mexico's capital city organized by grassroots activists who broadcasted their efforts on all social media platforms. These protests were followed by other collective action activities like federal building takeovers and glitter bombing, also broadcast online (Webb, 2023; Loyola-Hernández, 2025). Although the overall collective action for anti-femicide was clear, it was hard to understand what was going on, why it started, and what people were thinking. The way my social media bubble represented these events, they mainly called for the same changes, expressed similar grievances, and framed personal experiences connected to the movement from different points of view. As someone born and raised on the U.S. Mexican border, particularly in the town bordering the first recorded femicide in Mexico, my interests in this topic piqued and my own personal research on collective action was done. But, as participants of an online society, some find it difficult to step off these platforms and investigate further. For most, the information provided is all a viewer receives and can be fragmented or split between other personal interest posts populating the feed. In this case, I argue that there are ways to get a grasp on the narrative of a social media news cycle, particularly the anti-femicide social movement, by reviewing the narrative structure of its Instagram content and post metadata as a collection of topical images.

1.1 Research Problem

Recent reports show that more Americans consume their news and information from social media (Leppert 2024). Particularly, 26% of adults under 30 years old regularly get their news from visual based content social media platform, Tik Tok (Leppert, 2024). This does not include other platforms like Instagram, Facebook, or Twitter where 50% of adults get their news from social media overall (Pew Research, 2024). As many of these platforms provide content referred from user feedback for general personalization, users are not provided with full information about an event (Pew Research, 2020). Most of the time the information comes from sources that mirror a user's preferences and activity on the platform (Mosseri, 2021). These sources are user generated content that have been shown to portray different behaviors than what might be observed in offline and non-media contexts. The Instagram search algorithms also apply content recommendations through metrics to provide similar suggestions for users similar to each other (Sanchez-Moreno et al, 2020; Mosseri, 2021). This information bubble can provide a biased perspective of a real-time event that could be impactful to a community (Chen et al, 2020) and in turn enable a fragmented interpretation of what is happening. Similarly, each platform brings about different information consumption tendencies and so each provides a unique perspective of an event (Pelletier et al, 2020). These perspectives might be conflicting or fragmented, and information of a current event could be distorted.

Conflicting and fragmented information is not ideal for social movement practitioners, especially when they use social media to spread their cause. Current communication practices of social movements almost always require an online presence to connect with members and potential supporters (Peterson-Salahuddin, C.,2022). As real-time visual content has become a main force for informational content on social media, social movements might not have full control over their messaging and in-turn, lose sight of the online narrative (Leppert, 2024; Earl, Garrett, 2016; Moore-Gilbert et al, 2017). The anti-femicide movement in Mexico, for example, spreads across multiple online accounts and platforms, which initiate decentralized fragments of the movement's messaging goals (D'Ignazio, 2020). Much of the movement's content is a variation of informational, visual, call to action, or a combination, and can be